

2026-06-04 - ADC Reading

Goal

Use the nRF52840's built-in 12-bit SAR ADC (SAADC) to read an analog voltage, convert it to millivolts, and log the result periodically.

What you will learn

- How to declare an ADC channel in the Device Tree
- How to configure gain, reference, and acquisition time
- How to perform a single-shot ADC reading with `adc_read`
- How to convert raw counts to millivolts using `adc_raw_to_millivolts`
- Practical uses: battery monitoring, potentiometer, analog sensor

Overview

The nRF52840 SAADC has 8 channels, 8/10/12-bit resolution, configurable gain and reference. Zephyr abstracts this through the `adc` API.

app.overlay

```
/ {
    zephyr,user {
        /* AIN0 maps to P0.02 on nRF52840 */
        io-channels = <&adc 0>;
    };
};

&adc {
    status = "okay";
    #address-cells = <1>;
    #size-cells = <0>;

    channel@0 {
        reg = <0>;
        zephyr,gain = "ADC_GAIN_1_6";
        zephyr,reference = "ADC_REF_INTERNAL";
        zephyr,acquisition-time = <ADC_ACQ_TIME_DEFAULT>;
        zephyr,input-positive = <NRF_SAADC_AIN0>; /* P0.02 */
    };
};
```

prj.conf

```
CONFIG_ADC=y  
CONFIG_LOG=y
```

Reading the ADC

```
#include <zephyr/drivers/adc.h>  
  
#define ADC_NODE DT_PATH(zephyr_user)  
#define ADC_CHANNEL DT_IO_CHANNELS_INPUT_BY_IDX(ADC_NODE, 0)  
  
static const struct adc_dt_spec adc_ch = ADC_DT_SPEC_GET(ADC_NODE);  
  
void read_adc(void)  
{  
    int16_t raw;  
    struct adc_sequence seq = {  
        .buffer = &raw,  
        .buffer_size = sizeof(raw),  
    };  
  
    adc_sequence_init_dt(&adc_ch, &seq);  
    adc_read(adc_ch.dev, &seq);  
  
    int32_t mv = raw;  
    adc_raw_to_millivolts_dt(&adc_ch, &mv);  
    printk("ADC: raw=%d voltage=%d mV\n", raw, mv);  
}
```

nRF52840 SAADC AIN mapping

AIN GPIO Notes

AIN0 P0.02 Also used for UART RTS on DK — use with care

AIN1 P0.03 Also used for UART CTS

AIN2 P0.04 Free on most DK configs

AIN3 P0.05 Free on most DK configs

AIN4 P0.28

AIN5 P0.29

AIN6 P0.30

AIN7 P0.31

Measuring battery voltage (resistor divider)

Wire a 10 M Ω / 4.7 M Ω divider from VBAT to AIN, GND. Scale factor = $(10 + 4.7) / 4.7 \approx 3.13$.

```
int32_t vbat_mv = mv * 147 / 47;  /* fixed-point scale */
```

Why this matters

ADC is essential for battery monitoring, analog sensors (light, temperature via NTC, pressure), and potentiometer-based UIs. Understanding gain and reference settings prevents measurement errors.

Practice tasks

1. Wire a 10 k Ω potentiometer (wiper to AIN2/P0.04) and confirm readings sweep 0–3600 mV as you turn it.
2. Add a second ADC channel on AIN3 and log both simultaneously.
3. Average 8 consecutive readings to reduce noise.

Example folder

See `src/2026-06-04_ADC_Reading` for the day's code.

Next topic

Tomorrow we use PWM to smoothly dim an LED and control servo position.